

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

~ Patrick Lewis, Ethan Perez et al. (NIPS'20: Proceedings of the 34th International Conference on Neural Information Processing Systems)

Motivation

Find relevant evidence

Useful facts are scattered
across large collections.

Keep knowledge current

Update external sources
as facts change.

Inspect the sources

Retrieve passages that
a reader can examine.

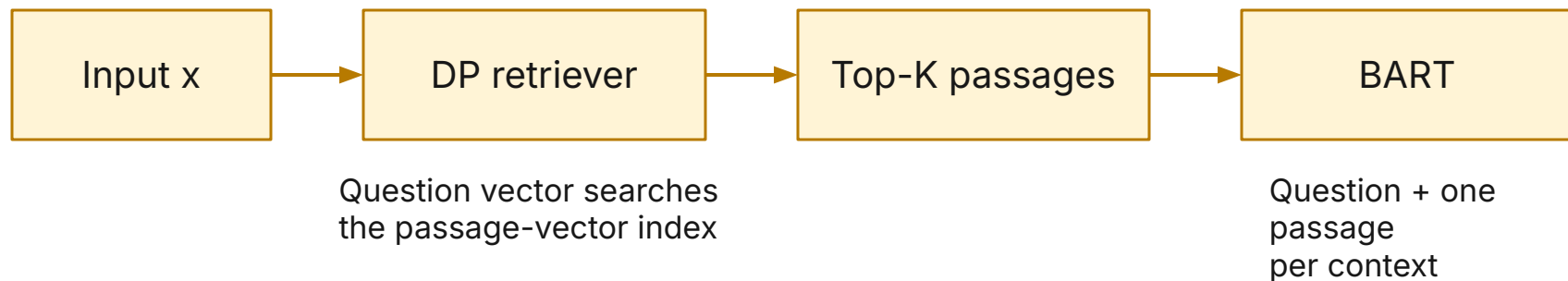
Use specific knowledge

Answer using information
from the chosen collection.

Key Vocabularies

Term	Meaning	Example
Parametric memory	Knowledge in learned weights	LLM's learned language and facts ingrained as model parameters
Non-parametric knowledge base	External searchable knowledge	Wikipedia passages + vector index
Retriever/ Dense passage retriever (DPR)	Model that ranks passages	DPR matches question and passage vectors
Dense embedding	A numerical vector representing a text's meaning	DPR converts questions and passages into vectors to score their relevance

Proposed Architecture



RAG-Sequence

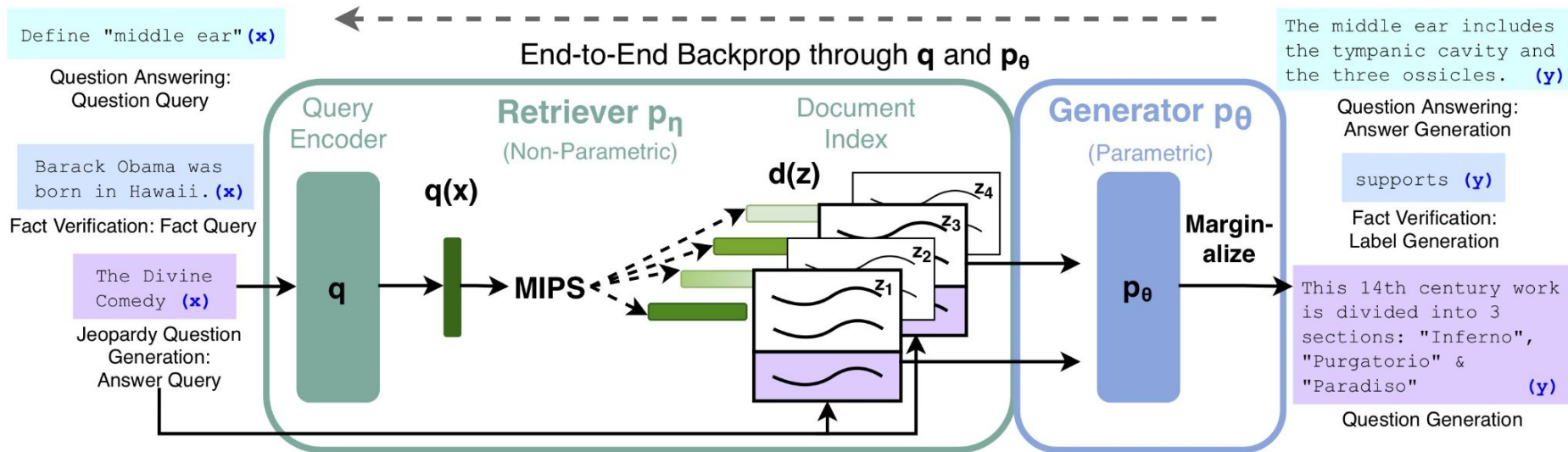
Weight and sum probabilities of the whole answer.

RAG-Token

Weight and sum next-token probabilities at each position.

Train query encoder + BART. Keep document encoder + index fixed.

Proposed Architecture



Experiment Design

Task	Benchmark	Why this task?	Metric
Factual QA	NQ, TriviaQA, WQ, CuratedTrec	Retrieve the correct fact	Exact match
Answer generation	MS MARCO	Write a full answer	ROUGE-L, BLEU-1
Clue generation	Jeopardy	Generate specific factual text	Q-BLEU-1, BLEU-1, human judgments
Fact verification	FEVER	Classify a claim using evidence	Label accuracy

Shared source: December 2018 Wikipedia, 21M passages of 100 words.
Train with $K = 5$ or 10 . Select test-time K on development data.

Results

Factual QA: exact match (%)

Model	NQ	TQA standard	TQA Wiki	WQ	CT
T5-11B	34.5	—	50.1	37.4	—
T5-11B + SSM	36.6	—	60.5	44.7	—
REALM	40.4	—	—	40.7	46.8
DPR	41.5	57.9	—	41.1	50.6
RAG-Token	44.1	55.2	66.1	45.5	50.0
RAG-Sequence	44.5	56.8	68.0	45.2	52.2

Generation and verification

Model	Jeop. BLEU-1	Jeop. Q-BLEU-1	MS MARCO ROUGE-L	MS MARCO BLEU-1	FEVER-3 Accuracy	FEVER-2 Accuracy
Prior best	—	—	49.8*	49.9*	76.8	92.2*
BART	15.1	19.7	38.2	41.6	64.0	81.1
RAG-Token	17.3	22.2	40.1	41.5	72.5	89.5
RAG-Sequence	14.7	21.4	40.8	44.2		

Higher = more correct answers/labels or greater reference overlap. *Gold context/evidence.

Extensions

Work	Direction	Key difference	Motivation / limitation
RAG (2020)	Generative baseline	DPR + BART; fixed top-K set for each input	Free-form outputs beyond extractive QA
FLARE (2023)	Timing	Retrieve during generation when a draft sentence is uncertain	Information needs change during long answers
Self-RAG (2024)	Judgment	Learn when to retrieve; critique passage relevance and support	Unnecessary retrieval and unsupported responses
HippoRAG (2024)	Structure	Links entities across passages in a knowledge graph and uses Personalized PageRank to retrieve relevant passages.	Independently ranking passages can miss connections needed for multi-hop question

FLARE adds timing. Self-RAG adds judgment. HippoRAG adds structure.

Strengths

Flexible, free-form answers

RAG can write an answer instead of selecting a text span. This supports tasks beyond extractive question answering.

Evidence improves generation

Retrieved passages help the generator use external facts. Gains across several tasks show that retrieval adds value beyond BART alone.

More factual, specific text

Human raters more often favored RAG's Jeopardy questions for factuality and specificity, providing evidence beyond word-overlap scores.

Weaknesses

Retrieval stays fixed

Later sentences may need new facts, but RAG keeps the initial passage set. FLARE responds by retrieving again during generation.

No explicit support check

An answer can still misrepresent its evidence. Self-RAG adds explicit judgments of passage relevance and whether the response is supported.

Difficulty connecting scattered evidence

RAG ranks passages independently, so it may miss relevant facts that become useful only when connected to another passage. HippoRAG addresses this by linking entities across passages and retrieving connected evidence through a knowledge graph.

Thank you!

Appendix (Visualization of the Problem)

My 25-page auto insurance policy

- I don't understand legal wordings
- I don't have proper attention span to read
- It's 25 PAGES!!

The problem

A disputed claim means finding the exact applicable clause across 25 pages.

A contract-based RAG system

Retrieve coverage and exclusions.
Explain the relevant conditions.
Show clause IDs and page numbers.

A low-ranked exclusion can still determine coverage. Check the original wording.